Embedding-based Neural Network Models for Book Recommendation in University Libraries

Jaeyoung Choi*, Chaeeun Han*, Heeyoon Yang*, Yeonkyoung Hong, Seoyoung Jeon, Yongjun Zhu**

Department of Library and Information Science, Sungkyunkwan University,

Seoul, South Korea

*Equal Contribution ** Corresponding Author

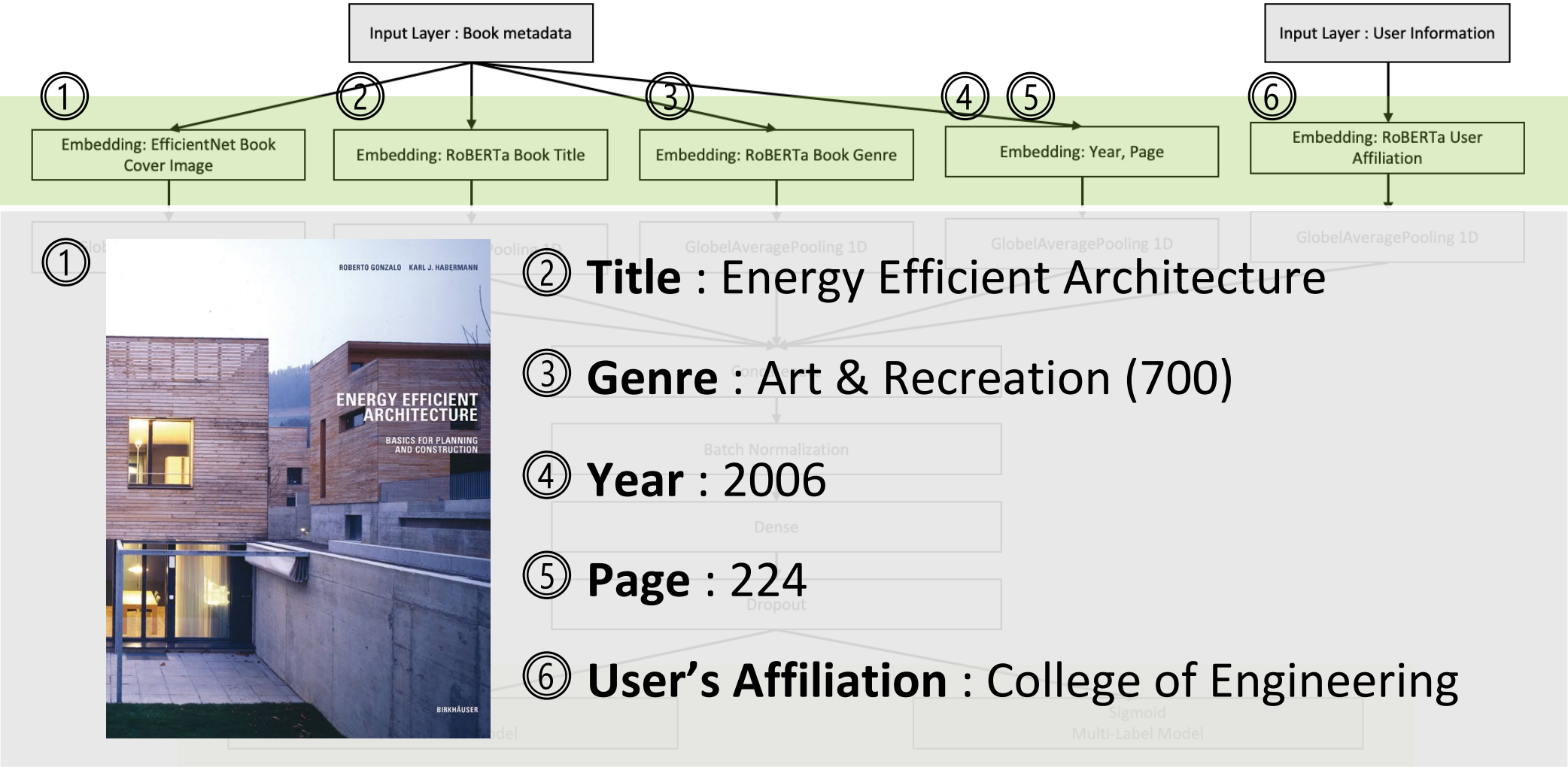
AI + Informetrics: Multi-disciplinary Interactions in the Era of Big Data Workshop at iConference 2021

Motivation

- **Library :** Burden in book recommendation due to vast collection
- **Student** : Difficult to choose books which include personal interest or present study field
- **Current book recommendation systems** : Require rating component

► **We propose personalized book recommendation models that only utilize book metadata and user data via embeddings.**

Model Architecture



② **Title** : Energy Efficient Architecture

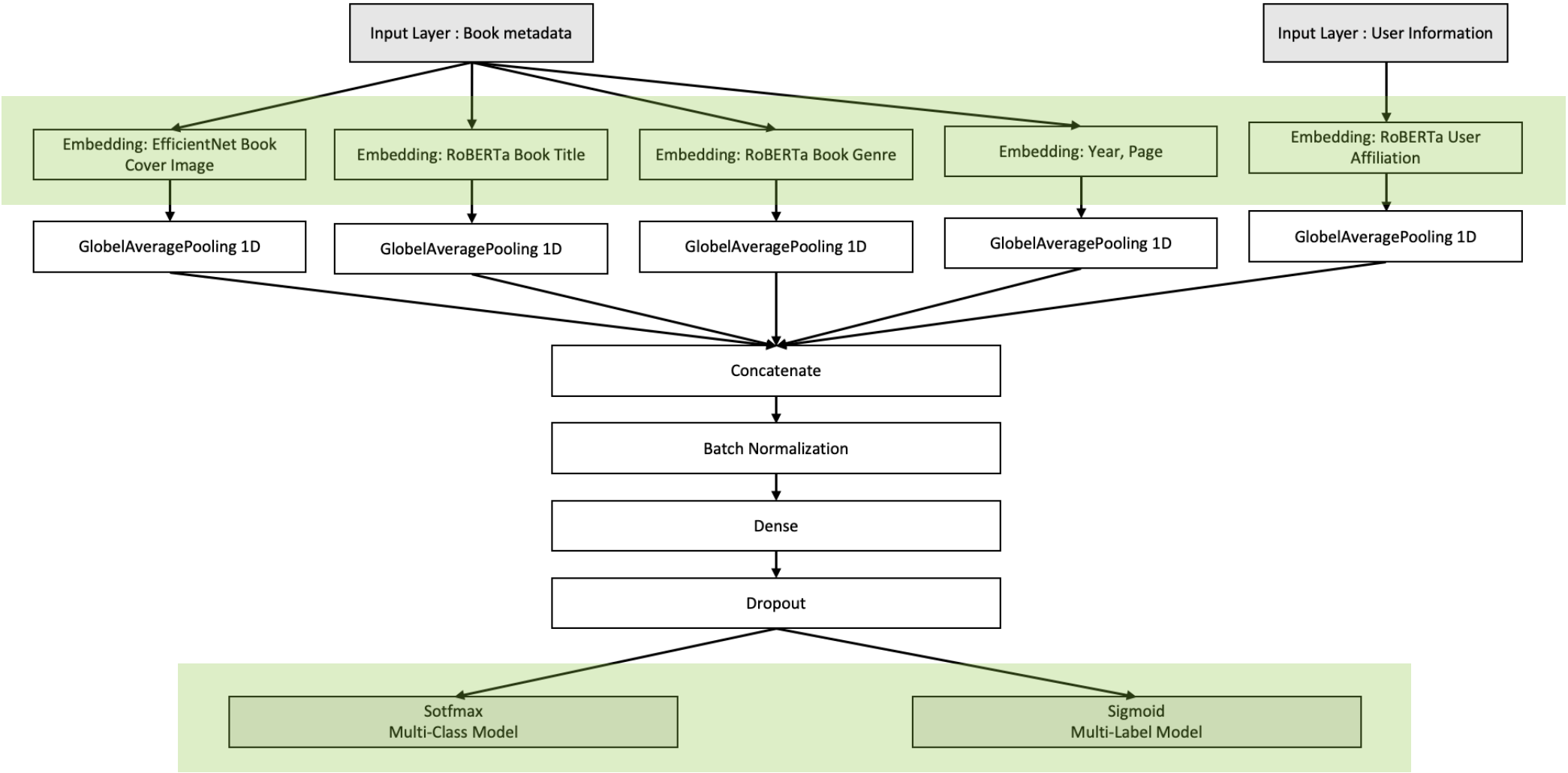
③ **Genre** : Art & Recreation (700)

④ **Year** : 2006

⑤ **Page** : 224

⑥ **User's Affiliation** : College of Engineering

Model Architecture

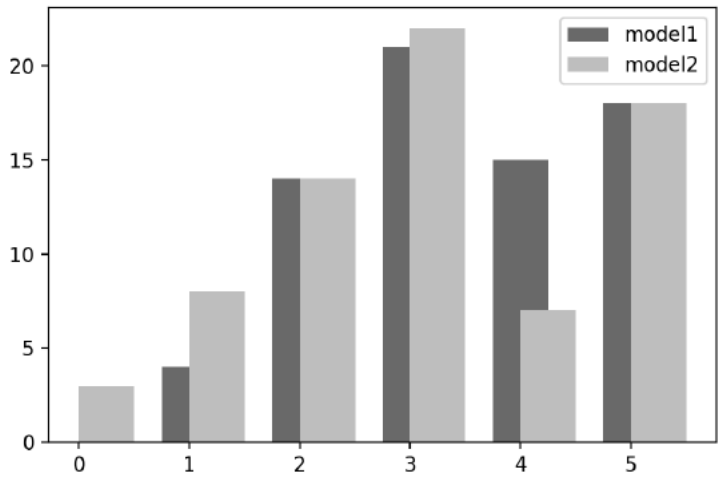


Results

Table 2. Accuracy for the Final Models

	multi-class model	multi-label model
accuracy	0.998	0.546

Fig. 2. Distribution of the User Evaluation Score



Trained multiple models by tuning hyperparameters

Constructed an ensemble through soft voting

Multi-class model : 3.4 average points (sd = 1.21)

Multi-label model : 3.0 average points (sd = 1.46)

Example of User Centered Evaluation

Book Input

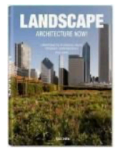
Beyond the horizon
The minimalist garden
(Small)landscape
Landscape design :urban furniture
Landscape space
(Magazine) LW =Landscape World
Process
Green urban landscapes
Process
Glass in structures :elements, concepts, designs
Masonry material and structure
Conceptual diagrams
Concept & competition
나의 한국 현대사 :1959-2014, 55년의 기록
그린 스튜디오 핸드북 :기본설계를 위한 환경적전략
Energy-efficient architecture :basics for planning and construction
Energy-efficiency upgrades :principles, details, examples

User Input

User's Affiliation : College of Engineering

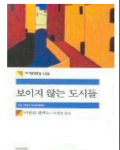
Model 1: Multi Class Model

등록번호	EM0916687	만족	불만족
서명	예술철학 :플라톤에서 들뢰즈까지		✓
저자	시릴 모라나, 에릭 우댕		
발행사항	서울 :미술문화, 2013		
형태사항	268 p. :삽화(일부천연색) ;21 cm		



등록번호	WW0153225	만족	불만족
서명	Landscape architecture now!	✓	
저자	Philip Jodidio.		
발행사항	Cologne :Taschen, 2012.		
형태사항	416 p		

등록번호	EM0923030	만족	불만족
서명	(지혜롭고 행복한) 집 한옥 :한옥의 과학과 미학	✓	
저자	임석재		
발행사항	서울 :인물과사상사, 2013		
형태사항	407 p.		



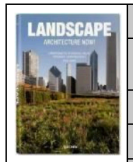
등록번호	EM0758068	만족	불만족
서명	보이지 않는 도시들	✓	
저자	이탈로 칼비노		
발행사항	서울 :민음사, 2007.		
형태사항	222 p.		

등록번호	EM0821732	만족	불만족
서명	약한건축	✓	
저자	쿠마 켄고		
발행사항	서울 :디자인하우스, 2009.		
형태사항	311 p.		

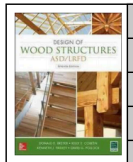
Model 2: Multi Label Model

등록번호	WW0145890	만족	불만족
서명	Architecture now! museums	✓	
저자	Philip Jodidio		
발행사항	Köln ;London :Taschen, c2010.		
형태사항	416 p. :col. ill. ;26 cm.		

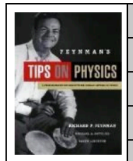
등록번호	WW0157825	만족	불만족
서명	Sustainable materials :with both eyes open	✓	
저자	Julian M.Allwood and Jonathan M. Cullen		
발행사항	Cambridge :UIT Cambridge, 2012		
형태사항	373 p. :col. ill., charts ;24 cm.		



등록번호	WW0153225	만족	불만족
서명	Landscape architecture now! :Landschafts Architektur heute!	✓	
저자	Philip Jodidio.		
발행사항	Cologne :Taschen, 2012.		
형태사항	416 p. :ill. (chiefly col.) ;26 cm.		



등록번호	WW0160207	만족	불만족
서명	Design of wood structures--ASD/LRFD	✓	
저자	Donald E. Breyer		
발행사항	New York :McGraw-Hill Education, c2015.		
형태사항	xxx, 976 p. :ill. ;25 cm.		



등록번호	WW0159414	만족	불만족
서명	Feynman's tips on physics	✓	
저자	Richard P. Feynman		
발행사항	New York :Basic Books, c2013.		
형태사항	xiv, 182 p. :Illustrations ;21 cm.		

Conclusion

- **Reliable performances for both models**

- **Higher feasibility in the real-world setting**

  ✓ Since the proposed model does not require ratings from users

  ✓ Able to apply in various domains other than university libraries

Thank you